

Deepwater Narrow-Window Drilling Protocol (illustrative)

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1. Purpose & Scope

This Standard Operating Procedure establishes operational protocols for safely traversing narrow drilling margins in deepwater Mahanadi offshore prospects (including MN-SM-DW-01, water depth 1,620 m). It specifies mandatory triggers for pre-emptive mud weight increases, drilling break management, and flow checks.

2. Responsibilities

- **Company Man / OIM**: Ultimate rig authority; enforces hold points and approves MOC submissions.
- **Drilling Superintendent (RTOC)**: Real-time surveillance lead; monitors LWD sonic/resistivity trends and authorizes weight-up.
- **Mud Engineer**: Formulates SOBM weighting schedule, supervises barite addition, and verifies mud density parity (in/out).
- **Driller**: Immediately executes ROP restriction upon encountering lithological breaks.

3. Definitions & Narrow Margin Context

- **Narrow Margin Drilling**: Intervals where the pore pressure (PP) and fracture gradient / casing shoe FIT margin is $\leq 0.65 \text{ ppg}$.
- **13-3/8 in Casing Shoe**: Set at 3,850 m MD with Formation Integrity Test (FIT) established at 12.10 ppg (1.45 SG).
- **Safe Drilling Envelope**: Maximum permissible Equivalent Circulating Density (ECD) must remain below 12.00 ppg to protect the 13-3/8 in shoe.

4. Operational Hold Points & Mandatory Triggers

1. Transition Zone Identification (4,150 m to 4,195 m MD):

- When compressional slowness (DT) increases above the Normal Compaction Trend (NCT) by $> 8 \text{ ms/ft}$, or resistivity (R_{deep}) departs below NCT, enter **High Alert Status**.
- Background gas trending upward from baseline requires continuous chromatograph inspection (SC_1 through SC_5).

2. Pre-emptive Weight-Up Trigger (4,172 m MD):

- In accordance with offset incident `INC-MN-DW-02-KICK-4195`, mud weight must NOT be kept at 11.20 ppg (1.34 SG) entering the pay sand.
- Mud weight must be raised pre-emptively from 11.20 ppg (1.34 SG) to **11.65 ppg (1.40 SG)** prior to crossing 4,195 m MD.
- This provides a minimum +0.17 ppg (+100 psi) overbalance against the projected sand pore pressure of 11.48 ppg (1.38 SG).

5. Drilling Break Protocol & ROP Cap

- **Sand Entry at 4,195 m MD**:
- Main reservoir pay (Miocene slope-channel turbidite sandstone) exhibits instantaneous ROP acceleration up to 34 m/hr.
- **Mandatory Action**: The driller must immediately throttle WOB/RPM to cap ROP at $\leq 12 \text{ m/hr}$.
- **Flow Check**: Perform a mandatory 15-minute static flow check at 4,196 m MD (pumps off, trip tank monitoring).
- Uncapped ROP in narrow windows generates heavy cuttings loading and annular friction surges that cause catastrophic losses (reference: `INC-MN-DW-03-LOSS-4222`).

6. Records & Regulatory References

- MOC records must be logged into RTOC digital ledger.
- References: ONGC Deepwater Drilling Manual §14; API RP 13B-1; IWCF Deepwater Well Control Compendium §4.